NeuroLearn AI

Milestone 1: Project Charter and Risk Register

Team: NeuroLearn AI (Group3)
Members: Annette Bazan and Natalia Solorzano
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Version: 0.1 | Review date: September 2, 2026
Repository: https://github.com/w6150197/ITAI-4376_Capstone_generative-ai-classroom-support

DRAFT FOR TEAM REVIEW. School stakeholder confirmation is pending. Role owners are provisional; targets describe planned tests, not completed results.

1. Project purpose and operational problem

NeuroLearn AI aims to help elementary special education teachers prepare activities based on individual student support preferences and receive timely notifications when students request assistance or reach a predefined inactivity threshold.

Elementary special education teachers and classroom aides must support students with different communication needs, sensory preferences, and instructional support needs. Preparing activities that accommodate these differences and recognizing when individual students need assistance can place competing demands on staff time.

During independent activities, staff may divide their attention among several students. A student who has difficulty communicating a request for help may wait for assistance, while educators interrupt other responsibilities to respond. Anticipated operational costs include preparation time, interruptions to instruction, and time students spend waiting for support. These assumptions require educator verification; no baseline measurements have been collected.

2. Stakeholder identification and confirmation

Annette plans to contact two nearby elementary schools within Houston Independent School District (HISD) to request a meeting with a special education teacher, coordinator, or administrator. The goals are to verify the problem, review the proposed scope, and identify a person willing to serve as project owner.

School outreach has not yet occurred. No named classroom project owner, written confirmation, HISD participation, classroom access, or student data access has been secured. General conversations with a teacher's aide were about their work and are not being presented as project validation. Contact dates, findings, and confirmation will be added after outreach.

3. Operating environment

NeuroLearn AI is planned as an application for elementary-age autistic students, with a student learning interface and a dashboard for teachers and classroom aides. Parents and caregivers would provide questionnaire information for teacher review. During the capstone, team members will simulate these roles using fictional profiles.

The application should be easy to install and affordable to operate on existing classroom devices. The prototype will run on devices available to the team. Supported operating systems, installation method, hosting, storage, and the AI provider remain to be selected after compatibility and cost checks. An Apple-specific version is a possible future direction, not a semester requirement.

The system will connect the questionnaire, profile storage, generated activities and hints, student interactions, and teacher notifications. An external AI service may require internet access and incur usage costs. No connection to school records is planned. Constraints include device performance, internet reliability, installation restrictions, staff time, accessibility needs, and the team's $0 additional-spending target.

4. Semester scope and boundaries

Included in the prototype

The instructional scope is basic counting and single-digit addition. Core functions are a parent/caregiver questionnaire; teacher-reviewed fictional student profiles; generated activities and guided hints; teacher approval before content reaches the student view; help and pause controls; and a teacher dashboard with help-request and inactivity notifications.

The prototype will separate teacher and student views, with a basic teacher access mechanism to be specified during design. The presentation will use simple navigation and teacher-controlled settings. AI will adapt instructional wording and hints; ordinary application rules will handle timers, help events, pauses, and alerts.

A direct help request will notify the teacher without waiting for the inactivity timer. Inactivity alerts will report an observable event, not diagnose distress. Paused sessions will not generate inactivity alerts. Both members will test the workflow using fictional data and simulated classroom interactions.

Outside the semester scope

Excluded work includes real student data collection, classroom trials with children, diagnosis, therapy recommendations, automated educational placement, claims of improved learning outcomes, school-record integrations, commercial launch, and payment processing. Music generation, advanced activity monitoring, automated break schedules, and an external LED accessory are future enhancements. The current scope includes a manual pause control.

5. Budget and future sustainability

The target is $0 in additional out-of-pocket spending, using existing devices and free or institution-provided resources where available. The team will verify service limits and usage costs before selecting tools. Free operation is a goal, not an established result.

A future annual subscription of $50–$100 per teacher may be explored after evaluating usefulness, operating costs, student usage, and educator feedback. This is a preliminary pricing hypothesis, not a launch commitment. MagicSchool's individual annual plan provides one comparison point; the products differ [3].

6. Draft success criteria

These are acceptance targets for the prototype. Tests will use fixed fictional profiles and documented application versions, device specifications, network conditions, and test steps. The team will retain outputs, scoring records, and timestamps so another reviewer can inspect the evidence.

Criterion	Threshold	Test method
SC1: Profile persistence	10 of 10 profiles preserve every required field.	Enter 10 fictional questionnaires. Save and reopen each profile, then compare every required field against the original entries.
SC2: Content quality and adaptation	At least 18 of 20 generated activities pass every checklist item.	Use 10 fixed profiles and two assigned tasks per profile. Before generation, freeze a checklist for mathematical correctness, task alignment, and requested instruction format. Retain all 20 outputs, including failures, with prompts and model settings.
SC3: Teacher approval gate	0 unapproved items appear in 20 attempts. Approved items become available in all 20 trials.	Use 10 activities and 10 hints. Try displaying each before approval; verify it is blocked. Approve it and verify it is accessible in the student view.
SC4: Help notification	20 of 20 requests show the correct student identifier within 5 seconds.	Record help-button event and dashboard-display timestamps. Calculate elapsed time and check the identifier for each request.
SC5: Inactivity rule	10 of 10 inactive trials alert between 180 and 185 seconds. Zero alerts in 10 active and 10 paused trials.	Set the threshold to 180 seconds. Run each trial for 200 seconds. Inactive trials have no interaction; active trials interact every 60 seconds; paused trials stay paused. Verify timestamps, early alerts, and duplicates.

Interpretation of results

Success demonstrates the tested workflow and content controls. It does not prove improved learning outcomes for autistic children or establish an ability to detect emotional distress. Human review remains required even if the content-quality threshold is met.

Provisional installation target

After selecting a supported platform, evaluate whether an adult tester can install and launch the prototype within 15 minutes using written instructions without developer assistance. This target remains provisional until installation and device requirements are defined.

Evaluation responsibilities

Annette will provisionally coordinate content and alert tests; Natalia will provisionally coordinate installation and integration tests. Both members will execute and review tests. Final responsibilities will be agreed by the team.

7. Risk register - first version

All risks are open. Owners are provisional coordinators; both team members share mitigation work. Likelihood: Low = unlikely, Medium = plausible, High = likely without action. Impact: Low = minor rework, Medium = recoverable delay, High = core deliverable blocked or significant safety/privacy harm. Review weekly.

Risk / category	**Likelihood / impact / owner**	**Mitigation**
R1 - Technical Incorrect or unsuitable activities and hints.	Medium / High Annette	Limit tasks to counting and addition. Use constrained prompts and a fixed review rubric. Require teacher approval before display.
R2 - Stakeholder / data access Delayed educator contact leaves assumptions unverified.	High / High Annette	Contact two local HISD elementary schools. Seek a named owner and written feedback. Continue with fictional profiles; do not treat synthetic tests as stakeholder confirmation.
R3 - Equipment Devices cannot run the application reliably or installation is restricted.	Medium / High Natalia	Identify supported devices and test early. Check school restrictions before proposing use. Prepare a clearly labeled recorded demonstration as a fallback.
R4 - Schedule Feature growth prevents MVP completion.	High / High Natalia	Prioritize the profile-to-activity-to-alert workflow. Review milestones weekly and defer music, hardware, and integrations.
R5 - Team Availability changes delay work or handoffs.	Medium / High Natalia	Track tasks and progress in GitHub. Document setup and decisions. Review workload weekly and reassign essential tasks when necessary.
R6 - Budget / services Free resources are insufficient or service limits interrupt tests.	Medium / High Natalia	Verify limits before selection and track usage. Use saved sample outputs only as a labeled demo fallback; do not report them as successful live generation.
R7 - Privacy / safety Sensitive student data or credentials enter the prototype or repository.	Medium / High Annette	Use fictional profiles only and label entry screens. Keep secrets out of GitHub. Review uploads; remove accidental disclosures and address exposed credentials promptly.
R8 - Notifications Missed, incorrect, or repeated alerts reduce reliability.	Medium / High Annette	Test event timing and identifiers. Suppress inactivity alerts during pauses and duplicate alerts for the same unresolved event. Keep educators responsible for supervision.

Owner key: Annette = Annette Bazan; Natalia = Natalia Solorzano. Final ownership will be confirmed by the team.

8. Ethics and privacy screening

Potential harm and human oversight

Students could encounter incorrect content, frustrating or inaccessible activities, or exposure of sensitive information. Teachers could receive missed or excessive alerts, rely too heavily on generated content, or interpret inactivity as distress. Parents and caregivers could be harmed if confidential information were shared without authorization.

Teachers will review and approve activities and hints before student access. Help and pause controls will remain available; pausing will not count as failure or trigger inactivity alerts. Notifications will describe help requests or inactivity and leave interpretation to educators. The system will not diagnose conditions, predict meltdowns, recommend treatment, make placement decisions, or serve as an emergency monitor.

Individualization and accessibility

The design will not assume that all autistic students learn alike or that limited spoken communication means limited understanding. Preferences will guide presentation without permanently restricting a student to one format. Teachers will be able to update them. Design priorities include simple navigation, readable instructions, presentation controls, and avoiding flashing visuals or unexpected audio.

Data touched and protections

Capstone data will consist of fictional profiles and questionnaire responses, generated content, simulated interaction records, and test timestamps. Fictional profiles will not be copied from identifiable children. Educator discussions will concern general classroom processes and will not request identifiable student information.

The prototype will not collect real student names, birth dates, contact information, photographs, voice recordings, diagnoses, individualized education programs, or school records. API keys, passwords, and private stakeholder contact information will remain outside the public repository. AI requests will contain only the fictional information needed for testing. Provider retention and model-training policies will be reviewed before selection.

Regulatory screening and future use

Future use involving school education records would require review of FERPA requirements and school rules for disclosure and vendor access [1]. An online service collecting personal information from children under 13 may be subject to COPPA, including notice and consent requirements depending on the circumstances [2]. Applicable state laws, district policies, provider terms, and institutional requirements also need review before deployment.

No claim of legal compliance or readiness for real-child use is made by this prototype. Any proposed student participation or real-data collection would require separate review with the professor and school, including required permissions and consent arrangements. A future real-data version would need documented authorized access, secure storage and transmission, retention, deletion, and incident-response procedures.

9. Team roles and shared responsibilities

Both members contribute to development, testing, planning, and documentation. The following assignments identify proposed coordination owners only. They are not finalized and do not restrict either member from contributing across roles.

Required role	Provisional owner	Coordination responsibility
Systems and Integration Lead	Natalia Solorzano	Coordinate questionnaire, profiles, student interface, dashboard, and integration.
Data and Model Lead	Annette Bazan	Define fictional profile data, prompts, personalization, and content evaluation.
Deployment and Infrastructure Lead	Natalia Solorzano	Coordinate supported devices, dependencies, configuration, hosting, and demo setup.
Governance Safety and Documentation Lead	Annette Bazan	Coordinate risk updates, privacy/safety requirements, stakeholder records, and documentation.

10. Confidentiality, IP, and external outreach

The professor facilitated a brief phone conversation between the team and Don Henderson, whom the team understands to be an Apple employee. He requested a synopsis. The professor then emailed the synopsis and copied the team. The team is awaiting further feedback. Dates and the email record can be added when available.

No formal sponsorship, Apple partnership, endorsement, or development-support commitment has been established. No written confidentiality or intellectual property agreement has been reported through this outreach. If collaboration develops, the team will document ownership, licensing, confidentiality, and permitted sharing under applicable institutional policies. An Apple-specific release remains a potential future direction.

This exploratory contact is separate from the required classroom problem-owner confirmation. No external sponsor is currently confirmed; sponsor-specific written terms are therefore not applicable at this stage.

11. Prior-work declaration

NeuroLearn AI is a new capstone project developed by Annette Bazan and Natalia Solorzano. It does not build on an earlier capstone or reuse work from a previous course or project.

At the time of this draft, Annette has created the GitHub repository and initial README and uploaded the preliminary planning outline. These are planning artifacts for this capstone. No application code, trained models, or functioning system components have been implemented. The team will add and test the proposed application during this semester. ChatGPT assistance with planning and this charter is disclosed below.

12. AI use statement

Review note: The statement below describes assistance with this deliverable..

I used ChatGPT to help organize and draft the NeuroLearn AI project charter, formulate proposed success criteria and a first-version risk register, and check wording, spelling, and grammar. ChatGPT also helped locate official student-privacy guidance and a pricing comparison and assemble the review document. Project details were supplied by me. AI did not provide stakeholder confirmation or generate reported implementation or test results. I am responsible for reviewing the final wording, verifying its accuracy, and ensuring it reflects our team's actual work and plans.

13. Stakeholder confirmation - pending attachment

No confirmation is attached to this review draft. A planned meeting or synopsis request does not replace the required email, signed form, or written statement from the project owner. After contact, attach the owner's actual response and complete the record below.

Confirmation record	**To be completed after contact**
Project owner and title	[Name and role]
Organization	[HISD elementary school or other confirmed organization]
Contact method and date	[Actual method and date]
Problem verified and feedback	[Accurate summary of what the owner said]
Scope reviewed	[Features or constraints discussed]
Evidence attached	[Email, signed form, or written statement and date]

14. Review items before submission

Confirm the team name and role owners; add the repository URL; verify all statements with Natalia; obtain and attach stakeholder confirmation; update the operational problem with educator feedback; and complete the final AI-use review. If confirmation is unavailable by the deadline, ask the professor how to handle the missing requirement rather than marking it complete.

*Required final filename: M01_NeuroLearn AI_AnnetteBazan_ITAI4376.pdf

References

[1] U.S. Department of Education. Education Technology Vendors. https://studentprivacy.ed.gov/audience/education-technology-vendors

[2] Federal Trade Commission. Complying with COPPA: Frequently Asked Questions. https://www.ftc.gov/business-guidance/resources/complying-coppa-frequently-asked-questions

[3] MagicSchool. Pricing & Plans. https://www.magicschool.ai/pricing (comparison reviewed September 1, 2026; individual annual plan listed at $99.96).